Kyobin Choo

Curriculum Vitae (Last updated: Sep. 16, 2026)

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Research Interests

Computer Vision, Medical imaging, Diffusion-like Models, Medical Image-to-Image Translation, Image Registration, Video Generation, Volumetric Image Segmentation, PET Quantification, Weakly Supervised Semantic Segmentation

Education

Yonsei University — Seoul, Republic of Korea
4th Year Ph.D. Student *in Computer Science* — *Mar. 2023 – Feb. 2028 (Expected)*
Advisor : Prof. Seong Jae Hwang
GPA : 4.24 / 4.3

Yonsei University — Seoul, Republic of Korea
B.S. *in Computer Science* — *Mar. 2017 – Feb. 2023*
GPA: 4.05 / 4.3 (Major: 4.21), 4 / 92 in class
2 years of absence for mandatory military service

Conference Publications *(* Equal contribution)*

[C1] **Kyobin Choo**, Y. Kim, H. Han, G. Park, C. Kim, S. Jung, S. J. Hwang, "QWERTY: Training-Free Motion Control via Query-Warped Video Diffusion Transformers", *European Conference on Computer Vision (ECCV), 2026* [PDF]
[C2] S. Jung*, J. Park*, Y. Choi, **Kyobin Choo**, M. H. Yang, S. J. Hwang, "Controlling Motion Transfer in Diffusion Transformers via Attention Heads", *European Conference on Computer Vision (ECCV), 2026* [PDF]
[C3] **Kyobin Choo**, H. Han, J. Kim, C. Yoon, S. J. Hwang, "Mono-Modalizing Extremely Heterogeneous Multi-Modal Medical Image Registration", *Medical Image Computing and Computer Assisted Intervention (MICCAI), 2025* [PDF]
[C4] D. Kim, S. Seo, B. Kim, **Kyobin Choo**, Y. Jun, M. Yun, "PRGNN: Pyramid Region Graph Neural Network for Region-Based Brain PET Classification", *Medical Image Computing and Computer Assisted Intervention (MICCAI), 2025*
[C5] Y. Jun, J. Park, **Kyobin Choo**, T. E. Choi, S. J. Hwang, "Disentangling Disentangled Representations: Towards Improved Latent Units via Diffusion Models", *Winter Conference on Applications of Computer Vision (WACV), 2025* [PDF]
[C6] **Kyobin Choo**, Y. Jun, M. Yun, S. J. Hwang, "Slice-Consistent 3D Volumetric Brain CT-to-MRI Translation with 2D Brownian Bridge Diffusion Model", *Medical Image Computing and Computer Assisted Intervention (MICCAI), 2024*, Early Accepted (Acceptance Rate $\leq 11\%$) [PDF]
[C7] **Kyobin Choo**, J. Jung, C. Seo, Y. Jun, Y. Kim, K. Myung, D. Y. Kim, S. J. Hwang, "안드로겐성탈모 진단 보조를 위한 헤어라인 이미지 분류 및 모발 굵기 측정 자동화 프레임워크", *Image Processing and Image Understanding (IPIU), 2024*

Journal Publications *(* Equal contribution)*

[J1] K. T. Oh, S. Lee, D. Kim, **Kyobin Choo**, S. Seo, Y. J. Yoon, Y. J. Park, Y. Lee, S. K. Yoo and M. Yun, "Prediction of FLAIR MRI from 18F-FDG PET/CT for the Evaluation of White Matter Hyperintensity Using Generative Adversarial Network", *Journal of Imaging Informatics in Medicine, 2026* (Impact Factor = 3.8)
[J2] D. Kim, G. Y. Kim, S. Seo, **Kyobin Choo**, S. Lee, H. Han, D. Kim, H. Lim, Y. Yun, S. Kang and M. Yun, "The Potency of DCLGAN for Reducing Scan time in PET/CT imaging with different radiotracers of clinical importance", *American Journal of Neuroradiology (AJNR), 2026* (Impact Factor = 3.7)
[J3] Y. J. Yoon, S. Seo, S. Lee, H. Lim, **Kyobin Choo**, D. Kim, H. Han, M. So, H. Kang, S. Kang, D. Kim, Y. G. Lee, T. J. Jeon, M. Yun, "Centiloid Values from Deep Learning-based CT Parcellation: A Valid Alternative to FreeSurfer", *Alzheimer's Research & Therapy, 2025* (Impact Factor = 7.6)

[J4] H. Han, **Kyobin Choo**, T. J. Jeon, S. Lee, S. Seo, D. Kim, S. J. Kim, S. H. Lee, M. Yun "GAN-Based Denoising for Scan Time Reduction and Motion Correction of F-18 FP-CIT PET/CT: A Multi-Center External Validation Study", *Clinical Nuclear Medicine, 2025* (Impact Factor = 10.0) [PDF]
[J5] S. Seo, Y. J. Yoon, S. Lee, H. Lim, **Kyobin Choo**, D. Kim, H. Han, S. Kang, J. Park, P. H. Lee, D. Kim, M. Yun "Striatal Dopamine Transporter Uptake Predicts Neuronal Hypometabolism and Visuospatial Function in Parkinson's Disease", *European Journal of Nuclear Medicine and Molecular Imaging, 2025* (Impact Factor = 8.6) [PDF]
[J6] D. Kim*, **Kyobin Choo***, S. Lee, S. Kang, M. Yun, J. Yang "Automated Quantification of Brain PET in PET/CT Using Deep Learning-Based CT-to-MR Translation: A Feasibility Study", *European Journal of Nuclear Medicine and Molecular Imaging, 2025* (Impact Factor = 8.6) [PDF]
[J7] **Kyobin Choo**, J. Joo, S. Lee, D. Kim, H. Lim, D. Kim, S. Kang, S. J. Hwang, M. Yun "Deep Learning-Based Pre-contrast CT Parcellation for MRI-Free Brain Amyloid PET Quantification", *Clinical Nuclear Medicine, 2025* (Impact Factor = 10.0) [PDF]
[J8] W. Han, S. Kang, **Kyobin Choo**, S. J. Hwang, "Complementary Branch Fusing Class and Semantic Knowledge for Robust Weakly Supervised Semantic Segmentation", *Pattern Recognition, 2024* (Impact Factor = 7.5) [PDF]

Conference Abstracts *(* Equal contribution)*

[A1] H. Han, S. Lee, **Kyobin Choo**, D. Kim, H. Lim, D. Kim, S. B. Seo, Y. Yoon, M. Yun,"Denoising for 1/4 Count PET Scans with Dose and Scan Time Reduction", *American College of Nuclear Medicine (ACNM), Annual Meeting, 2025*
[A2] D. Kim, **Kyobin Choo**, M. Yun, J. Yang,"CT-based deep learning framework for analyzing brain PET/CT", *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detector Conference (IEEE NSS MIC RTSD), 2024*
[A3] **Kyobin Choo**, J. Joo, D. Kim, S. Kang, S. J. Hwang, M. Yun,"MRI-free Brain Amyloid PET Quantification through Deep Learning-based Precontrast CT Parcellation", *Society of Nuclear Medicine and Molecular Imaging (SNMMI), 2024*
[A4] H. Han, **Kyobin Choo**, S. Kang, H. Lim, M. Yun,"Scan-time Reduction of Brain 18F-FP-CIT PET using DCLGAN", *Society of Nuclear Medicine and Molecular Imaging (SNMMI), 2024*
[A5] D. Kim*, **Kyobin Choo***, S. Kim, J. Yang, M. Yun,"Direct Deep Learning-based CT-to-MR Translation for Automated Quantitative Analysis in Brain PET/CT", *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room Temperature Semiconductor Detector Conference (IEEE NSS MIC RTSD), 2024*
[A6] D. Kim, S. Lee, D. Kim, **Kyobin Choo**, H. Lim, H. Han, J. Lee, M. Yun,"Deep-learning based multi-classification of FDG PET/CT images for the diagnosis of neurodegenerative diseases", *European Association of Nuclear Medicine (EANM), 2024*
[A7] D. Kim, **Kyobin Choo**, D. Kim, H. Lim, M. Yun,"Deep-learning based classification of dual-phase 18F-FP-CIT PET images for the diagnosis of Parkinsonism", *European Association of Nuclear Medicine (EANM), 2023*
[A8] S. Kim, **Kyobin Choo**, D. Kim, D. Kim, H. Lim, M. Yun,"Deep Learning-based CT to MRI Image Synthesis with Contrastive Loss", *Radiological Society of North America (RSNA), 2023*
[A9] D. Kim, **Kyobin Choo**, S. Kim, K. T. Oh, D. Kim, Y. J. Song, S. Kang, H. Lim, H. J. Park, M. Yun,"Correlation between 18F-Fluorodeoxyglucose, 18F-Florbetaben, and Mini-Mental State Examination score in patients with mild cognitive impairment due to Alzheimer's disease and AD dementia", *Society of Nuclear Medicine and Molecular Imaging (SNMMI), 2023*
[A10] D. Kim, K. T. Oh, J. H. Chun, **Kyobin Choo**, S. Kim, Y. J. Song, H. J. Park, J. Kim, S. Y. Kim, H. Lim, H. J. Park, M. Yun,"Prediction of IDH1 status in gliomas from 11C-Acetate PET/CT images using a convolutional neural network", *Society of Nuclear Medicine and Molecular Imaging (SNMMI), 2023*

Patents

[1] **Kyobin Choo**, Y. Jun, "Method, Apparatus and Computer Program for 3D Volume Data Conversion Using Multi-Slice Learned 2D Diffusion Models", 2025, Patent No. 102905764 (Currently under PCT application)

[2] **Kyobin Choo**, Y. Jun, "Method, Apparatus and Computer Program for Medical Image Conversion Using Style-Conditioned Diffusion Models", 2025, Patent No. 1028817520000 (Currently under PCT application)
[3] **Kyobin Choo**, "Method and System for Generating MRI Images based on CT Images", 2024, Patent No. 1026957180000
[4] **Kyobin Choo**, J. Lee, "Method and System for Brain Structure Segmentation for Parkinson's Disease Diagnosis", 2024, Patent No. 1026443420000
[5] **Kyobin Choo**, "Method and System for Quantitative Analysis of Brain Imaging based on CT Images", 2024, Patent No. 1026278740000 (Currently under PCT application)
[6] **Kyobin Choo**, "Method and System for Providing Information for Diagnosing Parkinson's Disease based on Machine Learning Model", 2024, Patent No. 1026443430000
[7] **Kyobin Choo**, "Method and System for Generating MRI Images based on CT Images", 2024, Patent No. 1026313890000
[8] **Kyobin Choo**, "Method and System for Generating High-quality Medical Image", 2023, Patent No. 1025936280000
[9] M. Yun, **Kyobin Choo**, "Method and System for Quantitative Analysis of Brain Imaging for Amyloid Measurement", 2023, Application No. 1020230084727

Research Experience

Medical Imaging & Computer Vision Lab @ Yonsei University Mar. 2022 – Present
Advisor: Prof. Seong Jae Hwang

- Training-free Video Motion Control
- Unsupervised Deformable Medical Image Registration
- Unsupervised Semantic Correspondence
- Volumetric Medical Image-to-Image Translation
- Disentanglement Representation Learning via Diffusion Models
- Weekly Supervised Semantic Segmentation

Nuclear Medicine & Artificial Intelligence Lab @ Yonsei University Dec. 2021 – Jan. 2025
Advisor: Prof. Mijin Yun

- PSMA Tumor Localization
- PET Denoising
- Brain CT & MRI Parcellation
- Brain CT-to-MRI Translation
- Brain PET Quantification
- PET/CT Classification for Alzheimer's Disease and Parkinson's Disease

Computational Intelligence & Photography Lab @ Yonsei University Mar. 2021 – Dec. 2021
Advisor: Prof. Seonjoo Kim

- Deepfake Detection

Accelerated Intelligent Systems Lab @ Yonsei University Dec. 2020 – Feb. 2021
Advisor: Prof. Jinho Lee

- Distributed Learning Acceleration
- FPGA Programming for Neural Network Acceleration

Academic Service

Conference Reviewer

- *Conference on Neural Information Processing Systems (NeurIPS), 2026*
- *Conference on Computer Vision and Pattern Recognition (CVPR), 2026*
- *European Conference on Computer Vision (ECCV), 2026*
- *Medical Image Computing and Computer Assisted Intervention (MICCAI), 2025 – 2026*
- *International Conference on Learning Representations (ICLR), 2025*
- *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2025*

Journal Reviewer

- *IEEE Transactions on Neural Networks and Learning Systems (TNNLS), 2025 (Impact Factor = 10.2)*
- *Pattern Recognition (PR), 2026 (Impact Factor = 7.6)*

Corporate Experience

AI/SW R&D Team Leader (CTO) @ Newcure M Inc. Jun. 2022 – Jan. 2025

- Patient-specific Brain PET Analysis Solution (Amyloid PET, 18F-FDG, 18F-FPCIT)
- CT-based Diagnosis Solutions for Alzheimer's Disease, Parkinson's Disease, and Vascular Dementia
- Short-time/Low-dose PET Scan Reconstruction Solution
- PSMA Tumor Detection Solution
- K-FDA Approvals for SW Medical Devices

Internship @ System LSI Division, Samsung Electronics Aug. 2021 – Dec. 2021

- CPU Scheduler Optimization for Application Processor (AP)

Project Experiences

MRI-Informed CT-based Neurological Prognostication after Cardiac Arrest Sep. 2026 – Feb. 2028

- MRI-informed Representation Learning from Paired CT-MRI Data
- CT-based Neurological Outcome Prediction with Pseudo-MRI Generation
- Cross-site Validation across Multiple Clinical Cohorts
- Funded by *National Research Foundation of Korea (NRF), Basic Research B*

Lumbar CT-to-MRI Translation for Spine Disease Diagnosis Support Jun. 2026 – Jun. 2027

- 3D Volumetric CT-to-MRI Translation using Flow Matching Models
- Image-to-Image Translation from Unregistered Image Pairs
- Cross-scanner Generalization and Domain Adaptation
- Industry-sponsored by *Vatech Co., Ltd.*

AI-assisted Causative Drug Diagnosis for Severe Cutaneous Adverse Reactions Dec. 2025 – Feb. 2028

- Cell-culture Image-based Prediction of Lymphocyte Transformation Test (LTT) Responses
- Multimodal Learning from PBMC Culture Images and LTT Measurements
- Collaborative Research with Severance Hospital, Yonsei University, and Seoul National University Hospital

Imaging & EMR-focused Multimodal Conversational Generative Model for Dementia Diagnosis Support Sep. 2024 – Dec. 2026

- Funded by *National Research Foundation of Korea (NRF)*

Androgenic Alopecia Classification and Hair Thickness Measurement Apr. 2023 – Aug. 2025

- Top-of-head Detection
- Hairline Image Segmentation and Classification
- Follicle and Hair Detection from Trichoscope Images
- Funded by *Seoul Business Agency of Korea (SBA)*

AI-based Software for Dementia Type Classification and Severity Diagnosis Apr. 2022 – Dec. 2024

- Brain CT Parcellation for PET/CT Quantification
- Short-time Brain PET Scan Denoising
- Brain Mid-sagittal Plane and AC-PC Line Extraction
- Brain T1-weighted MRI Generation from CT Scan
- Alzheimer's Disease Classification using 18F-FDG PET
- Funded by *National Research Foundation of Korea (NRF)*

Establishing a Reactive Glia-based Research Platform for Elucidating Dementia Pathogenesis Mar. 2021 – Dec. 2022

- Brain MRI Parcellation for PET Quantification
- Template-based Brain Amyloid PET Quantification
- Parkinson's Disease Diagnosis via 18F-FPCIT PET Quantification
- Funded by *National Research Foundation of Korea (NRF)*

Deepfake Detection Model Development Mar. 2020 – Dec. 2020

- Funded by *National Research Foundation of Korea (NRF)*

Skills

Programming

- Fluent in Python, Deep Learning Frameworks (Pytorch, Pytorch Lightning), Python Libraries (MONAI, Hugging Face, Scikit-Learn, etc), C++, Shell Script, LaTeX
- Git, Docker, AWS, Linux, AI Model Serving, Network Infrastructure
- DICOM protocol, PACS

Awards & Honors

AI Seoul Tech. Scholarship, Seoul Scholarship Foundation ***Selected 80 Ph.D. (national)***, Nov. 2025
Yonsei to Yonsei Scholarship (Outstanding GPA), Yonsei University Mar. 2023 – Sep. 2024
Excellence Award in Software Integrated Design, Yonsei University ***2nd / 35 teams***, Sep. 2021
Academic Excellence Award (Outstanding GPA), Yonsei University Sep. 2020
Academic Excellence Award (Outstanding GPA), Yonsei University Sep. 2018

Certifications

Medical Scientist Training Program, Yonsei University Jul. 2023
Long-term Internship, Samsung Electronics Dec. 2021

References

Seong Jae Hwang *(PhD Advisor)* Associate Professor at Yonsei University, Seoul, S.Korea
Mijin Yun Full Professor at Yonsei University College of Medicine, Seoul, S.Korea
Seonjoo Kim Full Professor at Yonsei University, Seoul, S.Korea
Jinho Lee Associate Professor at Seoul National University, Seoul, S.Korea